Supplementary Information for

Aggregate biocapacity does not identify its composition: a two-land model of ecological-capital drawdown

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Equation numbers without the prefix S refer to the main text.

S1 Symbols and model extensions

A global hectare (gha) is a hectare normalized to world-average biological productivity for the relevant accounting year. Annual Ecological Footprint and biocapacity account flows are expressed in gha, with the annual basis implicit in the account year (Global Footprint Network, 2009, 2019). In the model, B and E are annual account-equivalent quantities; in the empirical analysis, B and E_{prod} are observed NFA account values. The model contributions Y_f and Y_c and shortfall S are expressed in gha, but are not separately reported NFA series. The leaky debt state D is the time-integral of the model shortfall in gha yr; $\dot{D}$ and ηD are in gha.

Table S1: Symbols, meanings, units and roles. Per-hectare yields on the annual account basis are in gha ha^{-1}; $b_{G,c}$ is a regeneration-value coefficient in gha yr ha^{-1}; annual account quantities are in gha, with the accounting year implicit; the leaky debt state is in gha yr.

Symbol	Meaning	Unit	Role
A_f	fast provisioning land (cropland, intensive)	ha	state
A_c	ecological-capital land (forest, soil, wetland, long-rotation systems; effective class)	ha	state
A_r	reserve land, $A_{tot} - A_f - A_c$	ha	dependent balance
q	per-hectare capital-land quality index	dimensionless	state
P	human population	cap	state
D	model leaky-debt state (time-integral of model shortfall)	gha yr	state
b_f	fast-land account yield per hectare, $(b_{f0} + T_b)e^{-\alpha D}$	gha ha^{-1}	derived
$\tilde{b}_f(t)$	OWID/FAOSTAT cereal yield, a physical productivity sentinel (not all crops)	tonnes ha^{-1}	empirical input

Symbol	Meaning	Unit	Role
$\hat{b}_f(t)$	Base-anchored index-scaled proxy yield used only in SI S5	$\mathrm{gha\,ha^{-1}}$	proxy construction
b_{f0}	base fast-land yield per hectare	$\mathrm{gha\,ha^{-1}}$	parameter
b_c	capital-land direct yield	$\mathrm{gha\,ha^{-1}}$	parameter
$b_{G,c}$	capital regeneration-value coefficient	$\mathrm{gha\,yr\,ha^{-1}}$	parameter
$G_c(q)$	per-ha regeneration of quality, $\rho_c q(1-q/q_{\max})$	$\mathrm{yr^{-1}}$	rate function
$\Upsilon_c(q)$	per-ha capital value, $b_c + b_{G,c}G_c(q)$ (also written $b_{c,\mathrm{eff}}$)	$\mathrm{gha\,ha^{-1}}$	derived
Δb_{conv}	composition diagnostic, $b_f - \Upsilon_c(q)$	$\mathrm{gha\,ha^{-1}}$	derived
$Y_f,\ Y_c$	fast and capital yields, $b_f A_f$ and $\Upsilon_c A_c$	gha	derived
B	model biocapacity $Y_f + Y_c$; or NFA total under its account convention	gha	aggregate
E_{prod}	NFA Global Ecological Footprint of Production	gha	empirical account
E	model-defined footprint demand, eP (or exogenous frozen demand)	gha	aggregate
$X,\ C$	crop-proxy account $X = \hat{b}_f A_f$ and residual $C = B - X$; neither is physical capital	gha	illustrative accounting split
S	shortfall, $[E - \sigma_f Y_f - \sigma_c Y_c]_+$	gha	derived
u_c	conversion rate, $\min(S/b_f,\ A_c - A_c^{\min})/\tau_{\mathrm{conv}}$	$\mathrm{ha\,yr^{-1}}$	flow
K	carrying capacity, $\max(B/e, K_{\min})$	cap	algebraic
e	per-capita footprint	$\mathrm{gha\,cap^{-1}}$	parameter
r	intrinsic population growth rate	$\mathrm{yr^{-1}}$	parameter
ρ_c	quality regeneration rate	$\mathrm{yr^{-1}}$	parameter
$q_{\max}$	quality ceiling	dimensionless	parameter
μ	cropland maintenance rate (reserve→fast)	$\mathrm{yr^{-1}}$	parameter
η_f	fast-land retirement rate (fast→reserve)	$\mathrm{yr^{-1}}$	parameter
τ_{conv}	conversion time constant	yr	parameter
α	degradation rate of b_f per unit debt	$\mathrm{gha^{-1}\,yr^{-1}}$	parameter
η	debt-repayment rate	$\mathrm{yr^{-1}}$	parameter
$\tau_g,\ \tau_p$	regeneration and demographic delays	yr	parameter
$\sigma_f,\ \sigma_c$	human-available shares of the two flows	dimensionless	parameter
R_{rc}	surplus-gated restoration, $\chi_r A_r\,[B-E]_+$ (reserve→capital)	$\mathrm{ha\,yr^{-1}}$	gated flow
χ_r	restoration coefficient	$\mathrm{gha^{-1}\,yr^{-1}}$	parameter
T_b	technology yield increment, $\omega_{Tb}\Delta b/(1+e^{-\kappa_w(t-t_{wave})})$	$\mathrm{gha\,ha^{-1}}$	off except in SI S8
ω_{Tb}	activation flag for the technology wave	dimensionless	0 baseline; 1 in SI S8
Δb, κ_w, t_{wave}	amplitude, steepness and center of the wave	$\mathrm{gha\,ha^{-1}}$; $\mathrm{yr^{-1}}$; yr	active only in SI S8
$A_c^{\min}$, $A_r^{\min}$, $A_f^{\max}$, $K_{\min}$	capital viability floor, reserve floor, arable ceiling, carrying-capacity floor	ha, ha, ha, cap	typed floors
A_{tot}	total bioproductive area	ha	parameter

Symbol	Meaning	Unit	Role
k	$\mu/(\mu+\eta_f)$, fast share of non-capital land at equilibrium	dimensionless	derived
$B_{eq}(A_c)$	frozen-demand balance branch, $b_{f0}kA_{tot}-(b_{f0}k-b_c)A_c$	gha	derived
E_{ceil}	demand ceiling, $B_{eq}(A_c^{\min})$	gha	derived
$\lambda_{1,2}$	deficit-side Jacobian eigenvalues	yr^{-1}	derived
R_B, R_A, ψ_f	E/B, E/Y_f, Y_f/B (so $R_A = R_B/\psi_f$)	dimensionless	diagnostics
ϕ_0, w_X, w_C	base-year cropland share and log-mean weights	dimensionless	empirical decomposition
$s_X, \bar{s}_X, \bar{s}_C$	current and average fast-land shares	dimensionless	empirical decomposition
$\mathcal{R}_c(T)$	recovery metric, $(A_c(T)-A_c^{\text{deg}})/(A_c^{\text{init}}-A_c^{\text{deg}})$	dimensionless	derived
R_{fc}, χ	fast→capital restoration variant, $\chi A_f\,[B-E]_+$	$\text{ha}\,\text{yr}^{-1}$; $\text{gha}^{-1}\,\text{yr}^{-1}$	extension (off)
L_c, H_c	timber liquidation flow $[H_c-\sigma_c Y_c]_+/b_{G,c}$ and its target	$\text{ha}\,\text{yr}^{-1}$; gha	extension (off)

Model extensions. Timber liquidation and direct fast-to-capital restoration are not included in the reported simulations. Reservations of the human-available shares are considered as a policy extension in SI §S6. The technology wave is inactive in the baseline and gate-sign comparison and is activated only in the matched technology-wave experiment in SI §S8.

Conventions. The positive-part operator is exact, conversion proceeds through the time constant τ_{conv}, and land area is conserved among A_f, A_c, and A_r. The quality state q evolves separately from land area. The maintenance term μA_r permits a positive fast-land equilibrium. The reserve floor $A_r^{\min}$ and arable ceiling $A_f^{\max}$ are domain constraints; reported trajectories remain within them, with a minimum reserve of $0.80\,ha$ in the drawdown sweeps. The bounds $A_c \geq A_c^{\min}$, $A_f \geq 0$, $q \in [0, q_{\max}]$, and $K \geq K_{\min}$ are enforced.

S2 Derivations and proofs

S2.1 The composition identity with varying debt and quality

Differentiating $B = b_f A_f + \Upsilon_c(q)A_c$ with $b_f = (b_{f0}+T_b)e^{-\alpha D}$ and substituting Eqs. (6)–(7) gives the general identity

$$\dot{B} = \Delta b_{\text{conv}}\, u_c + b_f\big(\mu A_r - \eta_f A_f\big) + b_{G,c}G_c'(q)A_c\,\dot{q} + \Upsilon_c R_{rc} - \alpha\dot{D}\,b_f A_f + \frac{\dot{T}_b}{b_{f0}+T_b}\,b_f A_f. \quad \text{(S1)}$$

The conversion term has the sign of Δb_{conv}; the quality term has the sign of $G_c'(q)$ (non-negative for $q \leq q_{\max}/2$ and non-positive above it, since $\dot{q} \geq 0$). The debt term is non-positive while debt accrues, the technology term is non-negative while the wave rises, and the restoration term is non-negative. Proposition 1 in the main text assumes $T_b \equiv 0$, $R_{rc} \equiv 0$, and fixed D. In the demonstration parameterization, D evolves and reaches $0.20\,gha\,yr$ before decaying. At the initial state, the conversion contribution is $0.21\,\text{gha}\,\text{yr}^{-1}$, compared with $-0.005\,\text{gha}\,\text{yr}^{-1}$ from fast-land maintenance and retirement and a debt-term magnitude of approximately $0.004\,\text{gha}\,\text{yr}^{-1}$. Conversion therefore dominates the initial rate of change in aggregate biocapacity.

S2.2 The deficit-side affine system and its spectrum

On the deficit side with $q = q_{\max}$, D frozen, $R_{rc} = 0$ and $S/b_f < A_c - A_c^{\min}$, Eq. (5) gives $u_c = (E - b_f A_f - b_c A_c)/(b_f \tau_{\text{conv}})$, and with $A_r = A_{tot} - A_f - A_c$ Eqs. (6)–(7) become

$$\frac{d}{dt}\begin{pmatrix} A_f \\ A_c \end{pmatrix} = \begin{pmatrix} -\tau_{\text{conv}}^{-1} - \mu - \eta_f & -\mu - \dfrac{b_c}{b_f \tau_{\text{conv}}} \\ \tau_{\text{conv}}^{-1} & \dfrac{b_c}{b_f \tau_{\text{conv}}} \end{pmatrix} \begin{pmatrix} A_f \\ A_c \end{pmatrix} + \begin{pmatrix} \dfrac{E}{b_f \tau_{\text{conv}}} + \mu A_{tot} \\ -\dfrac{E}{b_f \tau_{\text{conv}}} \end{pmatrix}. \tag{S2}$$

Hence $\operatorname{tr} J = -\tau_{\text{conv}}^{-1}(1 - b_c/b_f) - \mu - \eta_f$ and $\det J = (\mu + \eta_f)(k - b_c/b_f)/\tau_{\text{conv}}$ with $k = \mu/(\mu + \eta_f)$. At the baseline $\operatorname{tr} J = -1.0512\,\text{yr}^{-1}$ and $\det J = 0.05353\,\text{yr}^{-2}$, and the eigenvalues are

$$\lambda_{1,2} = \tfrac{1}{2}\Big(\operatorname{tr} J \pm \sqrt{\operatorname{tr}^2 J - 4 \det J}\Big) = -0.0537\,\text{yr}^{-1},\ -0.9975\,\text{yr}^{-1}. \tag{S3}$$

Because J contains neither E nor the state, its spectrum is constant along the frozen-demand branch. At $E/E_{\text{ceil}} = 0.50, 0.90, 0.99, 0.999$, the slow eigenvalue is $\lambda_1 = -0.05366\,\text{yr}^{-1}$. It is independent of ρ_c (with $\lambda_1 = -0.0537\,\text{yr}^{-1}$ at $\rho_c = 0.04, 0.08, 0.16\,\text{yr}^{-1}$), varies modestly with τ_{conv} (-0.0553, -0.0537, -0.0505, $-0.0417\,\text{yr}^{-1}$ at $\tau_{\text{conv}} = 0.5, 1, 2, 5\,\text{yr}$), and varies more with μ (-0.0254, -0.0537, $-0.1099\,\text{yr}^{-1}$ at $\mu = 0.03, 0.06, 0.12\,\text{yr}^{-1}$). The slow eigenvalue remains negative throughout these parameter sweeps. The fixed point of the affine system is $A_c^*(E)$ of Eq. (19), with $A_f^* = k(A_{tot} - A_c^*)$; for $E > E_{\text{ceil}}$ it lies below $A_c^{\min}$, and the trajectory approaches the face $A_c = A_c^{\min}$ asymptotically, with limiting $u_c = 0$ and persistent shortfall $S^* = E - E_{\text{ceil}}$. At baseline, two members of the no-conversion equilibrium family are $(A_f^*, A_c^*, A_r^*, P^*) = (0.914\,\text{ha}, 0.824\,\text{ha}, 0.762\,\text{ha}, 1.488\,\text{cap})$ and $(1.091\,\text{ha}, 0.500\,\text{ha}, 0.909\,\text{ha}, 1.731\,\text{cap})$, both with $A_r^*/A_f^* = \eta_f/\mu = 0.8333$; the second is a sustainable cropland equilibrium supported by maintenance. The reserve-share mode $A_r/A_f \to \eta_f/\mu$ explains why the strict-inequality interval in the delayed-quality demonstration extends to approximately 149 yr, although most conversion is complete within 25 yr.

In the other branch of the minimum ($S/b_f \geq A_c - A_c^{\min}$), $u_c = (A_c - A_c^{\min})/\tau_{\text{conv}}$ and the distance to the floor obeys $\frac{d}{dt}(A_c - A_c^{\min}) = -(A_c - A_c^{\min})/\tau_{\text{conv}}$: it decays exponentially and never reaches zero in finite time. The floor is therefore approached asymptotically, and operational floor-proximity thresholds depend on the tolerance used to declare proximity (Section S4.2).

S2.3 Characteristic structure of the full delay system

With state $z = (A_f, A_c, q, P, D)$ the linearized system about any point of a given active set is $\dot{z}(t) = J_0 z(t) + J_g z(t - \tau_g) + J_p z(t - \tau_p)$ with characteristic equation $\det \Delta(\lambda) = 0$, $\Delta(\lambda) = \lambda I - J_0 - J_g e^{-\lambda \tau_g} - J_p e^{-\lambda \tau_p}$. J_0 carries the conversion and debt feedbacks on the current state (the 2×2 block above, plus the D row $\partial \dot{D}/\partial z = (-b_f, -\Upsilon_c, -b_{G,c} G_c' A_c, e, -\eta)$ on the deficit side and the $\partial b_f/\partial D = -\alpha b_f$ couplings); J_g has the single entry $G_c'(q^*)$ in the q row; J_p has the row $(rP^{*2}/K^{*2})\, e^{-1}\, \partial B/\partial z$ in the P position (from $\partial \dot{P}/\partial K = rP^2/K^2$ and $K = B/e$). Two scalar delayed modes decouple at the baseline. The quality mode linearized at $q_{\max}$ is $\dot{\xi} = -\rho_c \xi(t - \tau_g)$ (rate $-0.08\,\text{yr}^{-1}$ without the delay); its characteristic root satisfies $\lambda = -\rho_c e^{-\lambda \tau_g}$, and since $\rho_c \tau_g = 1.6 > \pi/2$ the linear delayed mode would be oscillatory. In the nonlinear model this cannot occur because $\dot{q} = G_c(q(t - \tau_g)) \geq 0$ for $q(t - \tau_g) \in [0, q_{\max}]$ and q is clamped at $q_{\max}$, so quality is monotone non-decreasing and converges to $q_{\max}$; the linear oscillation is an artifact of linearizing a one-sided law at its boundary. The population mode on the no-conversion face is the delayed logistic law with $r\tau_p = 0.5 < \pi/2$, below the Hutchinson threshold, so it is non-oscillatory at the baseline.

S2.4 Debt fixed points beyond the ceiling

For $E > E_{\text{ceil}}$ the limiting capital state is $A_c = A_c^{\min}$, approached asymptotically, and the persistent shortfall feeds the debt. With $B_{eq}(A_c^{\min}; D) = b_{f0} e^{-\alpha D} k(A_{tot} - A_c^{\min}) + b_c A_c^{\min}$, the debt

equation $\dot{D} = E - B_{eq}(A_c^{\min}; D) - \eta D$ has the fixed point $D^*(E)$ solving $\eta D = E - B_{eq}(A_c^{\min}; D)$; the right-hand side is increasing in D, so the fixed point is unique and attracting whenever $\eta > \alpha b_{f0} k(A_{tot} - A_c^{\min})$ ($0.034\,\mathrm{yr}^{-1} < 0.05\,\mathrm{yr}^{-1}$ at the baseline). At the baseline, $D^* = 1.186\,\mathrm{gha\,yr}$ at $E = 1.158\,\mathrm{gha}$ and $D^* = 5.375\,\mathrm{gha\,yr}$ at $E = 1.238\,\mathrm{gha}$, with $B(D^*) = 1.099$ and $0.969\,\mathrm{gha}$ and persistent shortfalls $S^* = 0.059$ and $0.269\,\mathrm{gha}$. Under frozen demand the debt therefore saturates; it is unbounded only if the shortfall is.

S2.5 Boundedness asymmetry and the crossover shortfall

From Eq. (12), $\ln b_f = \ln(b_{f0} + T_b) - \alpha D$. The first term lies in $[\ln b_{f0},\ \ln(b_{f0} + \Delta b)]$, so the technology channel contributes at most $\ln(1 + \Delta b/b_{f0}) = \ln(1.85/0.85) = 0.778$ to $\ln b_f$ at the baseline. From Eq. (13), $\dot{D} \le \bar{S} - \eta D$ with $\bar{S}$ the running supremum of the shortfall, so $D(t) \le \max\{D(0), \bar{S}/\eta\}$ and the degradation channel contributes at least $-\alpha\bar{S}/\eta$. The two are equal when $\bar{S} = S^\dagger = (\eta/\alpha)\ln(1 + \Delta b/b_{f0}) = 1.296\,\mathrm{gha}$ at the baseline. The fast-land contribution $Y_f = b_f A_f$ shares this structure; the capital-land contribution $\Upsilon_c A_c$ is unaffected by either channel. Under the additive law $b_f = b_{f0} + T_b - \alpha D$, both channels are bounded quantities of the same kind (Δb against αD^*) and the logarithmic asymmetry disappears, the bound above therefore applies to the multiplicative specification.

S2.6 Macro-ratio identities

Define $R_B = E/B$, $R_A = E/Y_f$ and the fast-land share $\psi_f = Y_f/B$; then $R_A = R_B/\psi_f$ is an identity. The direct-yield share within the capital-land contribution, $\psi_c = b_c A_c/Y_c$, is a different quantity and $R_A = R_B/\psi_c$ is false. $R_B = 1$ is aggregate balance. With baseline shares $\sigma_f = \sigma_c = 1$, $R_A > 1$ (demand exceeds the fast-land account yield) is necessary but not sufficient for conversion, because the shortfall requires $E > B = Y_f + Y_c$, not merely $E > Y_f$; with reserved flows the threshold changes. The capital-land area declines when $u_c > R_{rc}$, with u_c given by Eq. (5). Under the baseline shares, active conversion therefore requires $R_B > 1$; in the stated reservation extension $\sigma_f, \sigma_c < 1$, conversion and declining capital-land area can occur while $R_B \le 1$, so aggregate overshoot is not required in that extension. At the regeneration maximum $q^* = q_{\max}/2$ the direct-yield share of capital value is $b_c/(b_c + b_{G,c}\rho_c q_{\max}/4) = 0.758$ at the baseline; this is a property of that point, not a structural constant.

S3 Numerical methods and sensitivity analyses

Numerical method. The delay equations are integrated by the method of steps with an explicit first-order Euler scheme. The positive-part operator is exact; the state bounds are $A_c \ge A_c^{\min}$, $A_f \ge 0$, $q \in [0, q_{\max}]$, $P \ge 0$, $D \ge 0$, and $K \ge K_{\min}$. The reserve is calculated from $A_r = A_{tot} - A_f - A_c$. Pre-run histories of B and q are constant at their initial values over $[-\max(\tau_g, \tau_p), 0)$. In the technology-wave experiment, both runs use the common no-wave prehistory; the active $T_b(0)$ changes current $B(0)$, and subsequent wave-adjusted B values enter the delayed history used for $K(t - \tau_p)$. The demonstration run and bisection thresholds use $\Delta t = 0.05\,\mathrm{yr}$; drawdown-depth sweeps use $\Delta t = 0.1\,\mathrm{yr}$, with selected thresholds evaluated at $0.05\,\mathrm{yr}$. For $A_{c0} = 1.0\,\mathrm{ha}$, the $10^{-6}\,\mathrm{ha}$ floor-proximity threshold changes from 2.955 to 2.98 cap across these step sizes; threshold values are therefore reported as tolerance- and resolution-dependent bands.

Delayed versus current quality in the regeneration term. The reported trajectories use the delayed regeneration law $G_c(q(t-\tau_g))$, as specified in Eq. (10). A current-quality specification, $G_c(q(t))$, is evaluated as a sensitivity. Because q is monotone and the regenerative contribution is small ($b_{G,c}G_c \le 0.016\,\mathrm{gha\,ha}^{-1}$, compared with $b_c = 0.05\,\mathrm{gha\,ha}^{-1}$), the differences in the reported summaries are below 1%. The current-quality specification extends the demonstration

mask window by 1.3 yr and shifts the initial-population floor-proximity threshold by at most 0.001 cap (Table S2).

Table S2: Sensitivity to evaluating the regeneration flow at current rather than delayed quality (baseline parameter set, with $\tau_p = 25$ yr); reported trajectories use the delayed-quality specification. Biocapacity B, fast-land contribution Y_f, and capital-land contribution $\Upsilon_c(q)A_c$ are in gha; land areas A_c, A_f, A_{c0} are in ha; the operational initial-population floor-proximity threshold P_0 is in cap; delays and step intervals are in yr.

Quantity	current $q(t)$	delayed $q(t-\tau_g)$
Demonstration window with $\dot{B} > 0$, $\dot{A}_c < 0$ (yr)	150.3	149.0
Mask-end values B, A_c, A_f	0.825, 0.808, 0.923	0.825, 0.808, 0.923
Capital contribution $\Upsilon_c(q)A_c$ (initial → mask end, gha)	0.08364 → 0.04038	0.08364 → 0.04038
Fast-land contribution Y_f (initial → mask end, gha)	0.51000 → 0.78462	0.51000 → 0.78462
P_0 floor threshold ($A_{c0} = 1.0$ ha, $\tau_g = 5$ yr, $\Delta t = 0.1$ yr)	2.955	2.955
P_0 floor threshold ($A_{c0} = 1.0$ ha, $\tau_g = 20$ yr)	2.955	2.954
P_0 floor threshold ($A_{c0} = 1.0$ ha, $\tau_g = 40$ yr)	2.955	2.954

Well-posedness. The admissible state space $\Omega = \{(A_f, A_c, A_r, q, P, D) : A_i \geq 0,\ q \in [0, q_{\max}],\ A_f + A_c + A_r = A_{tot},\ K \geq K_{\min}\}$ is forward invariant under the stated bounds. The positive-part and minimum operators are Lipschitz, so the delay system has a unique solution for each admissible history. Stability at $S = 0$ is evaluated from the deficit side.

S4 Extended model results

S4.1 Fast-land yield sweep and parameter sensitivity of the composition diagnostic

Table S3 extends the fast-land yield sweep in Table 2(a) in the main text with the slope of the balance branch. The sign of Δb_{conv} remains positive as $b_{G,c}$ varies threefold: at $b_f = 0.85\,\text{gha}\,\text{ha}^{-1}$, $\Upsilon_c(q^*) = 0.058, 0.066, 0.074\,\text{gha}\,\text{ha}^{-1}$ for $b_{G,c} = 0.4, 0.8, 1.2\,\text{gha}\,\text{yr}\,\text{ha}^{-1}$, so $\Delta b_{\text{conv}} = +0.792, +0.784, +0.776\,\text{gha}\,\text{ha}^{-1}$.

Table S3: Fast-land yield sweep at the baseline (all other parameters fixed): composition diagnostic at q^*, eigenvalues of the deficit-side matrix, slope of the balance branch and demand ceiling.

b_f (gha ha^{-1})	$\Delta b_{\text{conv}}(q^*)$ (gha ha^{-1})	λ_1 (yr^{-1})	λ_2 (yr^{-1})	$b_f k - b_c$ (gha ha^{-1})	E_{ceil} (gha)
0.60	+0.534	−0.0522	−0.975	0.277	0.804
0.85	+0.784	−0.0537	−0.998	0.414	1.138
1.20	+1.134	−0.0547	−1.014	0.605	1.606
2.00	+1.934	−0.0556	−1.029	1.041	2.675

S4.2 Drawdown-depth ladder

Table S4 gives the minimum capital land reached during endogenous-demand runs ($A_{f0} = 0.6$, $q_0 = 0.9$, $\tau_g = 20$, $\tau_p = 25$ yr, $\chi_r = 0.10$, $\Delta t = 0.1$ yr, $T = 400$ yr) on a grid of initial population P_0; the full grid (step 0.1) is in `si_drawdown_depth.csv`. The depth is continuous and monotone in P_0 and in A_{c0}. The initial population P_0 (cap) at which the minimum comes within 10^{-2}, 10^{-3} and 10^{-4} ha of the floor is 1.8, 1.9, 2.1 ($A_{c0} = 0.40$ ha); 2.2, 2.3, 2.4 ($A_{c0} = 0.70$ ha); 2.6, 2.7, 2.8 ($A_{c0} = 1.00$ ha); 2.8, 2.9, 3.0 ($A_{c0} = 1.18$ ha); the bisection with a 10^{-6} ha floor-proximity tolerance ($\Delta t = 0.05$ yr, $T = 1000$ yr) gives 2.41, 2.70, 2.98, 3.14 cap. The thresholds are identical

to three decimals for $\chi_r = 0.10$ and $\chi_r = 0$ (2.387 and 2.955 cap at $A_{c0} = 0.40$ and 1.00 ha, $\Delta t = 0.1$ yr). For the delayed-quality specification at $A_{c0} = 1.00$ ha, the 10^{-6} ha threshold varies by at most 0.001 cap across $\tau_g = 5, 20, 40$ yr (Table S2); this small variation is specific to the tested range. The minimum is reached at the moment the shortfall closes ($B \geq E$ for the first time), which occurs between 3 and 25 yr after the start depending on P_0.

Table S4: Minimum capital land $\min_t A_c$ (ha) reached during the run, by initial capital A_{c0} (rows, ha) and initial population P_0 (columns, cap); typed floor $A_c^{\min} = 0.05$ ha.

$A_{c0} \setminus P_0$	1.0	1.5	2.0	2.5	2.8	3.0	3.5	4.0
0.4	0.396	0.147	0.050	0.050	0.050	0.050	0.050	0.050
0.7	0.700	0.442	0.121	0.050	0.050	0.050	0.050	0.050
1.0	1.000	0.731	0.398	0.075	0.050	0.050	0.050	0.050
1.18	1.180	0.898	0.557	0.230	0.055	0.050	0.050	0.050

S4.3 Recovery metric and the role of the restoration flow

The recovery metric is $\mathcal{R}_c(T) = (A_c(T) - A_c^{\text{deg}})/(A_c^{\text{init}} - A_c^{\text{deg}})$, with $\mathcal{R}_c = 0$ no recovery, 1 full recovery to the reference, < 0 continued decline and > 1 over-restoration. Table S5 evaluates it on the delayed-quality drawdown runs with $A_{c0} = 1.0$ ha ($A_c^{\text{init}} = 1.0$ ha, $A_c^{\text{deg}} = \min_t A_c$ in ha, $T = 400$ yr). Without restoration the capital-land area remains at its minimum because no return flow exists. With the surplus-gated flow, recovery is partial across these six initial-population cases ($\mathcal{R}_c = 0.21$–0.89); none reaches the reference. The distinct gate-sign experiment in SI §S4 tests other initial states and intervention settings, including cases of over-restoration.

Table S5: Recovery of the capital-land area after a demand-driven drawdown ($A_{c0} = 1.0$ ha, $A_{f0} = 0.6$ ha, $q_0 = 0.9$, $T = 400$ yr): initial population P_0 is in cap, initial footprint E_0 in gha, capital-land values in ha, and the recovery metric is dimensionless.

P_0	E_0	$\min_t A_c$	t(min) (yr)	$A_c(T)$, $\chi_r = 0$	$A_c(T)$, $\chi_r = 0.10$	$\mathcal{R}_c(T)$, $\chi_r = 0.10$
1.5	0.825	0.731	3.0	0.731	0.969	0.89
2.0	1.100	0.398	3.8	0.398	0.677	0.46
2.5	1.375	0.075	4.1	0.075	0.433	0.39
3.0	1.650	0.050	13.8	0.050	0.328	0.29
3.5	1.925	0.050	20.6	0.050	0.278	0.24
4.0	2.200	0.050	26.6	0.050	0.250	0.21

S4.4 Gate-sign phase dependence

The gate-sign comparison begins from a post-drawdown state ($A_c = 0.058$ ha near the floor, $A_f = 0.6$ ha, $q = 0.9$, and $P = 1.2$ cap). At a degraded equilibrium with $R_B = 1$, $B = E$, so neither the surplus-gated nor deficit-gated restoration flow is active. The reference equilibrium has $A_c^{\text{init}} = 0.8225$ ha and $P^{\text{init}} = 1.489$ cap; the degraded value is $A_c^{\text{deg}} = 0.0513$ ha. All rows use $T_b = 0$ and the stated initial state.

Without restoration, A_c remains at its minimum because the model has no return flow. Surplus-gated restoration yields partial or over-restoration depending on χ_r; at $\chi_r \geq 0.05$, the flow continues while $B > E$, and the supported population falls to 0.97–0.90 of its reference value. A target cap returns A_c to its reference value ($\mathcal{R}_c = 1.00$). Deficit-gated restoration acts during the shortfall, but at the tested rates its contribution is smaller than concurrent conversion and the recovery fraction remains near zero. Freezing conversion under surplus-gated restoration raises peak debt from 0.152 to 0.543 gha yr; restoration then reallocates reserve area to capital land while fast-land maintenance is reduced. These outcomes depend on the specified initial state

Table S6: Restoration gate and recovery outcomes at $T = 100$ and 900 yr. The table reports end-state land areas, population relative to the reference, trajectory extrema, and peak debt D (gha yr). Annual account values $B = E$ are in gha; land areas are in ha; population ratios and recovery metrics are dimensionless. Each row approaches $R_B = 1$ ($E = B$).

Policy / gate	$\mathcal{R}_c(100)$	$\mathcal{R}_c(900)$	A_c	A_f	A_r	$B = E$	P/P^{init}	$\min A_r$	$\max A_c$	D_{peak}
none ($R_{rc} = 0$)	−0.00	−0.00	0.051	1.336	1.113	1.138	1.39	1.113	0.057	0.152
surplus-gated, $\chi_r = 0.02$	0.69	0.73	0.612	1.030	0.858	0.906	1.11	0.858	0.612	0.152
surplus-gated, $\chi_r = 0.05$	1.08	1.07	0.875	0.886	0.738	0.797	0.97	0.724	0.885	0.152
surplus-gated, $\chi_r = 0.10$	1.28	1.27	1.030	0.802	0.668	0.733	0.90	0.602	1.092	0.152
surplus, $\chi_r = 0.10$, target cap $A_c^* = 0.8225$	1.00	1.00	0.822	0.915	0.763	0.819	1.00	0.664	0.822	0.152
surplus, $\chi_r = 0.10$, conversion frozen	1.35	1.36	1.097	0.765	0.638	0.705	0.86	0.602	1.097	0.543
deficit-gated, $\chi_r = 0.05$	0.00	0.00	0.055	1.334	1.112	1.136	1.39	1.112	0.060	0.145
deficit-gated, $\chi_r = 0.10$	0.01	0.01	0.059	1.331	1.110	1.135	1.39	1.110	0.065	0.139
deficit, $\chi_r = 0.10$, conversion frozen	0.05	0.05	0.088	1.315	1.096	1.123	1.37	1.096	0.088	0.159

and intervention rates. The gate-sign comparisons use $T_b = 0$; the controlled wave/no-wave debt comparison is reported in SI §S8.

S4.5 Regime table

Table S7: The composition diagnostic Δb_{conv} (gha ha^{-1}) as the decision condition for mask visibility. There is no interior area-based capacity maximum in the quality-based model; the only shape distinction is the sign of Δb_{conv}.

Regime	Condition	Behavior
composition-premium	$\Delta b_{\text{conv}} > 0$	the conversion contribution to accounting B is positive; net B also depends on other flows (the mask is possible in the baseline)
composition-neutral	$\Delta b_{\text{conv}} = 0$	first-order neutrality; non-generic
composition-discount	$\Delta b_{\text{conv}} < 0$	conversion lowers B at the conversion margin; conversion alone cannot produce a composition mask

S5 Empirical details

S5.1 Data and vintages

The empirical backbone is the National Ecological Footprint and Biocapacity Accounts (NFA), Edition 2026, results and metadata version 1.0 (Lo et al., 2026). Although the workbook covers 1961–2025, the primary analysis uses complete world observations for 1961–2023; the 2024–2025 estimated rows are excluded. The 1961–2022 sensitivity uses the same edition and population series. The release recalculates common historical years and reports biocapacity by fishing grounds, built-up land, cropland, grazing land, and forest biocapacity. These are account categories, not a common-boundary physical-stock inventory.

The workbook's `pop_1000_persons` field is UN PopStat population in thousands of persons; it is converted to persons before recomputing per-capita values. The field `efp_total_gha` is the Global Ecological Footprint of Production, denoted E_{prod} here. Per-capita values are ratios of the unrounded account totals and population, not the workbook's rounded per-capita columns. Table S8 summarizes endpoint changes. For 1961–2023, B is 9,725,417,562 to 12,142,843,961 gha, population is 3.06487 to 8.09173 billion, B/P is 3.17319 to 1.50065 gha person^{-1}, and E_{prod}/P is 2.19700 to 2.56462 gha person^{-1}. The identity $\Delta \ln B = \Delta \ln P + \Delta \ln(B/P)$ holds to rounding; it is an extensive/intensive accounting split, not a causal decomposition. The first year in which $E_{\text{prod}}/B > 1$ is 1973 in both windows.

Table S8: World endpoint changes in NFA Edition 2026. The 1961–2022 column is a same-edition sensitivity; 2024–2025 estimates are excluded. Population and per-capita quantities are recomputed from workbook totals.

Measure	1961–2023	1961–2022
$\Delta B/B_{1961}$	+24.8568%	+24.5450%
$\Delta \ln B$	+0.221997	+0.219497
$\Delta P/P_{1961}$	+164.0154%	+161.7211%
$\Delta \ln P$	+0.970837	+0.962109
$\Delta(B/P)/(B/P)_{1961}$	−52.7085%	−52.4131%
$\Delta \ln(B/P)$	−0.748840	−0.742612
$\Delta(E_{\text{prod}}/P)/(E_{\text{prod}}/P)_{1961}$	+16.7327%	+17.6994%
First year $E_{\text{prod}}/B > 1$	1973	1973

S5.2 Divisia decomposition, sensitivity to the base-year share and level fallback

The crop proxy is an illustrative accounting construction. Let A_f be the independently sourced OWID/FAOSTAT world cropland-area series (arable plus permanent-crop area, ha) and let $\tilde{b}_f$ be the independently sourced cereal-yield series (tonnes per hectare) (Our World in Data, 2026a,b). Cereal yield is a sentinel for crop productivity, not a measure of all-crop output. For a declared 1961 account-share anchor ϕ_0, define $\hat{b}_f(t) = \hat{b}_f(1961)\tilde{b}_f(t)/\tilde{b}_f(1961)$, where $\hat{b}_f(1961)A_f(1961) = \phi_0 B(1961)$, then $X_t = \hat{b}_f(t)A_f(t)$ and $C_t = B_t - X_t$. Thus X is an index-scaled account proxy and C is the residual required to recover the NFA aggregate; C is not observed ecological capital, non-cropland biocapacity, or the model quantity $\Upsilon_c(q)A_c$.

When X and C are positive at both endpoints, the exact log-mean Divisia identity is

$$\Delta \ln B = w_X\left(\Delta \ln \tilde{b}_f + \Delta \ln A_f\right) + w_C \Delta \ln C, \qquad w_X = \frac{L(X_T, X_0)}{L(B_T, B_0)}, \quad w_C = \frac{L(C_T, C_0)}{L(B_T, B_0)}, \tag{S4}$$

where $L(a, b) = (b - a)/(\ln b - \ln a)$ is the logarithmic mean, with its continuous limit at $a = b$. The measured NFA cropland share in 1961 is $\phi_0 = 0.1286472202$ and is used as the central anchor, but the mapping of that account share to the external area–yield proxy is an assumption, not an independently identified physical-capital observation. Table S9 reports the full selected sensitivity for 1961–2023. At the central anchor, the cereal-yield, area, and residual terms are $+0.268662$, $+0.037710$, and -0.084375, summing to $\Delta \ln B = +0.221997$. The same-edition 1961–2022 sensitivity gives $+0.264696$, $+0.037506$, and -0.082705, summing to $+0.219497$. The decomposition is exact for the defined proxy but its allocation depends on ϕ_0 and must not be read as a physical-stock attribution.

Table S9: Sensitivity of the proxy Divisia decomposition to the 1961 account-share anchor ϕ_0 (world, 1961–2023). Contributions and weights are dimensionless. The $\phi_0 = 0.128647$ row uses the measured NFA cropland share in 1961; other rows are illustrative calibrations.

ϕ_0	w_X	w_C	Cereal yield	Area	Residual C	Sum
0.100000	0.1833	0.7957	+0.2088	+0.0293	−0.0162	+0.2220
0.128647	0.2358	0.7352	+0.2687	+0.0377	−0.0844	+0.2220
0.150000	0.2749	0.6893	+0.3133	+0.0440	−0.1352	+0.2220
0.190000	0.3483	0.6008	+0.3968	+0.0557	−0.2305	+0.2220
0.250000	0.4582	0.4581	+0.5221	+0.0733	−0.3734	+0.2220
0.300000	0.5499	0.3178	+0.6265	+0.0879	−0.4924	+0.2220
0.320000	0.5866	0.2454	+0.6683	+0.0938	−0.5401	+0.2220
0.340000	0.6232	0.1009	+0.7100	+0.0997	−0.5877	+0.2220

For $\phi_0 \geq 0.340531$ in 2023 (threshold 0.343400 for 2022), the residual C is nonpositive at the terminal endpoint and logarithmic weighting is invalid. In that range the exact level identity $\Delta B = \Delta X + \Delta C$ is used instead; it remains an accounting identity and carries no stock interpretation. The selected anchor has $X_{2023}/B_{2023} = 0.377784$ and a positive residual, but the proxy fast-land share and residual contribution vary across the sensitivity values. These calculations use the same NFA B in all rows; the residual absorbs the difference between the index-scaled crop proxy and that aggregate, including account-category, boundary, valuation, vintage, and calibration differences.

S5.3 Cropland-account calibration

Direct NFA cropland-account movements are reported separately from the illustrative proxy. From 1961 to 2023, NFA cropland biocapacity rises 2.870055-fold while NFA cropland area rises 1.198029-fold; the quotient of account biocapacity by NFA area rises 2.395647-fold. For the same-edition 1961–2022 sensitivity the ratios are 2.851441, 1.198326, and 2.379520. The per-hectare

quantity is an NFA account-value quotient, not an independent physical crop-yield measure. By comparison, the external OWID/FAOSTAT area series rises 1.173414-fold and cereal yield rises 3.124667-fold over 1961–2023. Those independent series provide a sentinel proxy, not a reconstruction of the NFA cropland account or a measure of all-crops output. The model's representative $b_{f0} = 0.85\,\mathrm{gha\,ha^{-1}}$ remains a conditional parameter, not an empirical fit to these account ratios.

S5.4 Observation-operator hierarchy

- *NFA accounting observations.* Total and component biocapacity values in gha support statements about the composition and movement of the NFA accounts under the stated edition and conventions. They do not directly observe biomass, soil condition, wetland extent, fish biomass, or the model state A_c.

- *Fast-provisioning area.* The external world cropland-area series is arable plus permanent-crop area, 1961–2023 (ha) (Our World in Data, 2026a). It is used as an independently sourced proxy input, not treated as identical to the NFA cropland-area series.

- *Crop-productivity sentinel.* The OWID/FAOSTAT cereal-yield series, 1961–2024, is measured in tonnes per hectare (Our World in Data, 2026b). It is not an all-crops index and is not converted into a physical stock. The base-anchored $\hat{b}_f$ is only the index-scaled term in the illustrative account proxy.

- *Ecological-capital components.* Forest extent, growing stock or biomass, soil carbon and condition, wetland extent and condition, and species- or stock-specific fish biomass could inform separate physical observation operators. They would require harmonized spatial boundaries, reference years, definitions, uncertainty, and explicit treatment of overlap. No common-boundary global physical-stock series for the relevant components was assembled here. Existing NFA forest biocapacity and the residual C are not substitutes for those measurements.

- *Model mapping.* In the model, $Y_c = \Upsilon_c(q)A_c$ is a model-defined capital contribution. Empirically, neither A_c nor q is identified by B or by an NFA component; the proxy residual $C = B - X$ is not assumed to equal Y_c. A defensible observation operator would need independent physical measures and a declared mapping before a model-to-data comparison.

  Table S10 summarizes this mapping without equating an NFA category to a model state.

S5.5 Regeneration anchors

The capital-land regeneration timescale is a lumped recruitment-response timescale. Plot-level age–above-ground-biomass records from Neotropical secondary forests (2ndFOR; 1,334 plots, 41 chronosequences, 35 good fits) yield median $t_{50} \approx 30\,\mathrm{yr}$ and $\tau_r \approx 38\,\mathrm{yr}$ under a lagged saturating recovery model (Poorter et al., 2016). Deplete-then-rebuild trajectories for 33 collapsed fish stocks in RAM Legacy (B/B_{MSY}) yield $t_{50} \approx 5.6\,\mathrm{yr}$ and $\tau_r \approx 3.4\,\mathrm{yr}$ (Hutchings and Reynolds, 2004; Neubauer et al., 2013); soil organic carbon re-equilibrates over approximately 17–23 yr following land-use change (Poeplau et al., 2011). Forest biomass reaches about half of old-growth values by 20 yr and 90% at a median of about 66 yr, while approximately 29% of collapsed fish stocks recover to 50% within 5–15 yr. These records span distinct systems and are not pooled into a single empirical rate. Forest and soil observations provide decadal-scale context for the representative effective response scale of about 29 yr; the 10–40 yr interval is a modeling range for the lumped capital-land class, not a pooled empirical estimate. The faster fish-stock episodes are not treated as a direct lower bound on that class. Neither ρ_c nor the delay τ_g is calibrated from these observations.

Table S10: Conceptual mapping from two-land model quantities to NFA information and independent proxy inputs. The rows specify possible observation operators, not identities between NFA categories and model states. Annual account values are in gha; land areas are in ha.

Model quantity	Closest account or proxy information	Interpretation and identification limit
Fast area and proxy account A_f, $X = \hat{b}_f A_f$	NFA cropland account; independently sourced OWID cropland area and cereal yield	NFA cropland biocapacity is an account value; the external area–yield product is an illustrative, base-anchored proxy. Neither is an observed model fast-land yield or a one-to-one physical measure.
Capital area, quality and contribution A_c, q, $Y_c = \Upsilon_c(q)A_c$	No direct NFA counterpart; the proxy residual $C = B - X$ is defined by subtraction	In the model $Y_c = \Upsilon_c(q)A_c$; empirically C is not assumed equal to Y_c. It also absorbs differences in boundaries, account categories (including built-up land), valuation, vintage and proxy calibration, and is not a physical-stock measure.
Reserve area A_r	No direct NFA account counterpart	A_r closes the model's conserved-area ledger; it is not recoverable from the NFA aggregate.

Table S11: Observability ladder: what is directly measured, what is identifiable by construction, and what remains unidentified.

Tier	Quantities	Status
Observable physical area input	OWID cropland area [ha]; other mapped land-cover classes where available	Measured extent, but mapping to effective A_c is not direct
Observable population	P [persons]	Direct in the UN PopStat workbook field after unit conversion
Observable productivity sentinel	Cereal yield [tonnes ha^{-1}]	Physical yield of cereals only; not all-crops output or ecological capital
Observable annual account value	NFA B, E_{prod} and components [gha]	Measured only under the NFA accounting convention; value-weighted accounts
Proxy account and residual	$X = \hat{b}_f A_f$ and $C = B - X$ [gha]	Identified by construction for the stated inputs and anchor; not physical stocks and not assumed equal to $\Upsilon_c(q)A_c$
Effective capital contribution	$Y_c = \Upsilon_c(q)A_c$ [gha]	A model quantity; no common-boundary physical observation operator is assembled here
Not separately identifiable	A_c, q, b_c, $b_{G,c}$, ρ_c and D	Requires independent physical-state, quality, regeneration or degradation observations and a declared mapping

S5.6 Recursive identifiability in the land-type accounts

NFA Edition 2026 shows why the aggregate should be read together with its account composition, without translating account movement into physical-stock movement. From 1961 to 2023, cropland contributes +24.0577 percentage points of the +24.8568% change in total B; built-up land contributes +7.4416 points, forest biocapacity −5.9525, grazing land −0.2688, and fishing grounds −0.4212. The non-cropland account sum, which includes built-up land, changes by only +0.7990 points (1.0092×) because those unlike account movements net against one another. In the same-edition 1961–2022 sensitivity, the corresponding contributions are cropland +23.8183, built-up land +7.2854, forest biocapacity −5.8137, grazing land −0.2890, and fishing grounds −0.4559 points; the non-cropland sum changes by +0.7267 points (1.0083×). These are accounting attributions, not evidence that forest biomass, soil condition, wetland extent, fish biomass, or modeled A_c declined. Built-up land is an NFA account class, not ecological capital; its inclusion in the non-cropland sum is arithmetic, not ecological aggregation.

S5.7 Physical-stock scoreboard: specification, not inventory

No common-boundary global physical-stock series for the relevant components was assembled, and no physical-stock composite or global decline finding is presented. This absence does not establish that such a series could never be assembled. Table S12 is an unpopulated specification of candidate measures and comparability requirements; it contains no empirical endpoint values and is not a model calibration.

Heterogeneous physical stocks cannot be integrated by simply summing their native units: forest biomass or soil carbon (mass), wetland or cropland extent (area), fish biomass by species or assessed stock, freshwater storage (volume), and condition indicators (often dimensionless or categorical) are neither dimensionally commensurate nor necessarily spatially disjoint. Annual flows such as harvest, catch, increment, recharge, or emissions must also remain distinct from stocks and extents. Even after normalization into a common score, overlap, baselines, uncertainty, ecological thresholds, valuation, and permitted substitution determine the interpretation. Component measures therefore form a nonadditive vector unless common spatial and temporal boundaries, normalization or valuation, and substitution rules are declared. A scalar composite is not neutral and cannot be inferred from the NFA gha accounts. This distinction is consistent with ecosystem-extent and ecosystem-condition accounts being reported separately from ecosystem-service and monetary presentations in SEEA Ecosystem Accounting (United Nations et al., 2021).

In the two-land model, A_c is an effective area state (ha) and q is a separate capacity-quality index. Neither is directly observed in the NFA aggregate. A physical-stock assessment could inform a model observation operator only after its boundaries and mapping to those effective states are specified. The proxy residual $C = B - X = B - \hat{b}_f A_f$ is not ecological capital, a measured non-cropland stock, or an estimate of $\Upsilon_c(q)A_c$; the NFA components remain account results rather than physical-stock observations.

Table S12: Physical-stock scoreboard specification. The rows are unpopulated: no common-boundary global physical-stock series for the relevant components or endpoint values were assembled. Stocks, extents, condition measures, and annual flows are not combined into a composite.

Component	Physical stock, extent, or condition to define	Annual flow or pressure (separate)	Comparability rule required
Forests	Extent (ha); growing stock or biomass (m^3 or tonnes C); stated condition	Net annual increment, harvest, or deforestation	Common forest definition and mapped area; reference year; treatment of plantations and overlap
Soils	Soil organic-carbon stock or density for a stated depth; structure or erosion condition	Carbon accumulation or loss; erosion where measured	Consistent depth, bulk density, land cover, and sampling method
Wetlands	Extent (ha) and an explicit ecological-condition measure	Carbon accumulation, water-storage change, or nutrient flux	Wetland definition and overlap with forest and other land-cover classes
Fisheries	Spawning-stock biomass by species or stock (tonnes); age or size structure	Recruitment, catch, and landings	Stock identity, assessment method, reference point, and uncertainty
Freshwater	Surface-water or groundwater storage (m^3), level, and quality	Recharge and withdrawal	Basin or aquifer boundary; separate stored volume from annual flow

S6 Policy extensions

Reserved biocapacity share. With a common reservation share $\sigma_f = \sigma_c = \sigma < 1$, human-available supply is σB and the shortfall is $S = [E - \sigma B]_+$. The reservation lowers the demand at which shortfall-driven conversion begins; conversion is inactive while $E \leq \sigma B$. With distinct shares, the corresponding condition is $E \leq \sigma_f Y_f + \sigma_c Y_c$, which need not equal σB. These are conditional results for the specified parameterization and do not establish robustness to disturbances or parameter uncertainty.

Typed floors and conversion stop. Reserve $A_r \geq A_r^{\min}$, capital viability $A_c \geq A_c^{\min}$, arable ceiling $A_f \leq A_f^{\max}$, and $K \geq K_{\min}$ are separate constraints. A conversion stop triggered directly by $A_c = A_c^{\min}$ preserves the capital-land floor. An A_f ceiling constrains fast-land extent but does not by itself guarantee that floor unless the stop rule is linked to the capital floor and remaining-area constraints. In Eq. (5), the conversion rate tends to zero as $A_c \to A_c^{\min}$. An operator based only on aggregate B cannot guarantee the typed capital-land floor because B does not identify A_c.

Institutional signature. Related work distinguishes protective from mobilizing institutional feedback (Abaee, 2026a). In this model, surplus-gated restoration is inactive during a shortfall, whereas deficit-gated restoration acts during it and can reduce provision further (Section S4.4).

Aggregation and typed feasibility. In related assessment work, B is the linear, perfectly substitutable member of an aggregator family (Lemma A of Abaee, 2026b). Aggregate feasibility can therefore coexist with violation of a typed floor on A_c; no positive-elasticity aggregator is uniformly safe (Theorem S2(ii) in Abaee (2026b)). Under pathwise feasibility, a decline below a typed floor during the transition remains a violation even if the endpoint later recovers. These results distinguish an aggregate account criterion from constraints on individual components.

S7 The one-stock comparator

This one-stock formulation is a structural comparator, not a reduction of the two-land model. Its state A has comparator-specific stock units and is not the two-land state A_c. Let

$$
\begin{aligned}
&G(A) = \rho A\left(1 - \frac{A}{A_{\max}}\right), \qquad B(t) = b(t)A(t) + b_G G(A(t)), \qquad E(t) = eP(t);\\
&\dot{A}(t) = G\big(A(t-\tau_g)\big) - \frac{[E(t) - \sigma b(t)A(t)]_+}{b_G};\\
&\dot{P}(t) = rP(t)\left(1 - \frac{P(t)}{K(t-\tau_p)}\right), \qquad K(t) = \max\left(\frac{B(t)}{e}, K_{\min}\right);\\
&\dot{D}(t) = [E(t) - B(t)]_+ - \eta D(t), \qquad b(t) = \big(b_0 + T_b(t)\big)e^{-\alpha D(t)};\\
&T_b(t) = \omega_{Tb}\frac{\Delta b}{1 + \exp[-\kappa(t - t_w)]}.
\end{aligned}
$$

Here σ is a dimensionless human-available share and $\omega_{Tb} \in \{0, 1\}$ switches the logistic technology increment on or off. A, $A_{\max}$, and the imposed lower bound A_{ext} use comparator-specific stock units; $A_{\max}$ is the logistic growth scale, not an imposed upper bound, and $G(A)$ is stock-unit yr^{-1}. The coefficients b, b_0, T_b, and Δb are gha per stock unit, while b_G is gha yr per stock unit, so both terms in B are annual account values in gha. E and B use the GFN/NFA convention, with the annual basis implicit; P, K, and $K_{\min}$ are in cap, and e is gha cap^{-1}. The leaky state D is in gha yr, so $\dot{D}$ and ηD are in gha; α is gha^{-1} yr^{-1}. Rates ρ, r, η, κ are in yr^{-1}, and delays τ_g, τ_p and the wave midpoint t_w are in yr. The numerical implementation imposes $A \geq A_{\text{ext}}$, $P, D \geq 0$, and $K \geq K_{\min}$ and uses the exact positive-part operator. Delayed histories are constant at their initial values; RK4 integration uses linear interpolation of delayed states. The figure steps are 0.1 yr for the full trajectory (checked at 0.05 and 0.025 yr), 0.01 yr for the fixed-demand mask (with a 0.02 yr shortfall sweep), 0.2 yr for the delay pair (checked at 0.1 yr), and 0.5 yr for the finite-grid classification.

With fixed b, $T_b = 0$, $D = 0$, and $\sigma = 1$, interior equilibria with $0 < A < A_{\max}$ and $B(A)/e > K_{\min}$ satisfy $E = B(A)$ and $P = B(A)/e$, forming a one-parameter family. The corresponding curve is

$$
B(A) = (b + b_G\rho)A - \frac{b_G\rho}{A_{\max}}A^2, \qquad B'(A) = b + b_G\rho - 2b_G\rho\frac{A}{A_{\max}}.
$$

An interior maximum on $0 < A < A_{\max}$ requires $b_G\rho > b$. In the debt-free constant-parameter set used for the delay and basin illustrations, $b = 0.5$, $b_G = 0.8$, and $\rho = 0.05$, so $b_G\rho = 0.04 < b$ and $B(A)$ is increasing on $0 < A < A_{\max}$. In the full-trajectory illustration, by contrast, $b(t)$ varies with the technology wave and debt; no monotonicity or equilibrium-fold claim is made for that time-varying case.

At the debt-free reference equilibrium $A^* = 0.8$ for the constant-parameter set $(b, b_G, \rho, A_{\max}, e, r) = (0.5, 0.8, 0.05, 1.2, 0.55, 0.02)$, $B^* = 0.4107$ gha and $P^* = B^*/e = 0.7467$ cap. The characteristic equation has a neutral root tangent to the equilibrium family and a positive real root. The latter is $0.5883\,\text{yr}^{-1}$ at $(\tau_g, \tau_p) = (0, 0)$ and $0.6250\,\text{yr}^{-1}$ at $(30, 25)$ yr. These are comparator-specific local results, not a Hopf classification or a stability claim for the two-land system.

Figures S1–S4 show conditional simulations of this comparator. In Fig. S3, the same initial state approaches $A_{\max}$ for $\tau_g = 10$ yr and reaches the imposed floor for $\tau_g = 30$ yr when $\tau_p = 25$ yr; this pair illustrates dependence on the regeneration delay for one initial condition, not a universal critical delay. Figure S4 classifies a finite grid of initial conditions with τ_p held fixed. The proportions of grid cells are not probabilities over initial conditions or measures of a continuous basin.

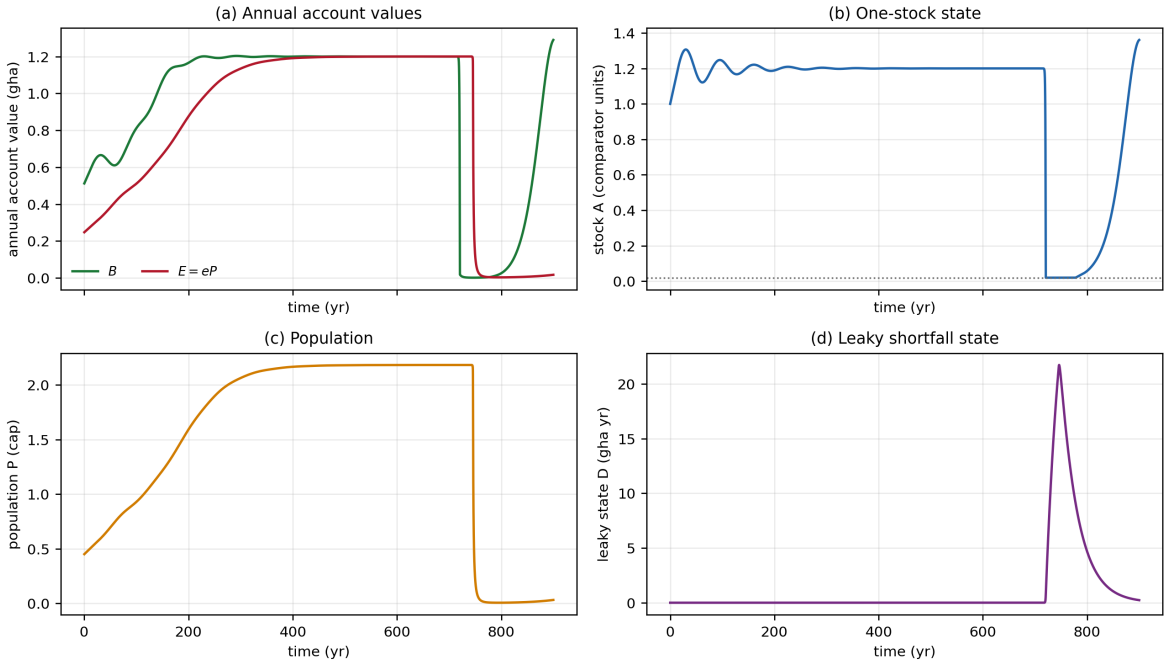


Figure S1: Full delayed one-stock comparator with a logistic technology increment. Parameters: $A_0 = 1.0$ stock unit, $P_0 = 0.45\,\mathrm{cap}$, $b_0 = 0.5$, $b_G = 0.6$, $\rho = 0.08\,\mathrm{yr}^{-1}$, $A_{\max} = 1.2$ stock units, $e = 0.55\,\mathrm{gha\,cap}^{-1}$, $r = 0.02\,\mathrm{yr}^{-1}$, $\eta = 0.03\,\mathrm{yr}^{-1}$, $\alpha = 0.2\,\mathrm{gha}^{-1}\,\mathrm{yr}^{-1}$, $\tau_g = 15\,\mathrm{yr}$, $\tau_p = 25\,\mathrm{yr}$, $A_{\mathrm{ext}} = 0.02$ stock units, $K_{\min} = 10^{-4}\,\mathrm{cap}$, $\Delta b = 0.5\,\mathrm{gha}$ per stock unit, $\kappa = 0.04\,\mathrm{yr}^{-1}$, and $t_w = 120\,\mathrm{yr}$; $\sigma = 1$, $\omega_{Tb} = 1$, and $D_0 = 0$. The positive-part operator is exact. B and E are in gha, A in comparator-specific stock units, P in cap, and D in gha yr. The trajectory reaches the imposed lower bound near 721 yr and subsequently recovers as modeled demand falls; the bound is not absorbing. This is a conditional model illustration, not an observation or a two-land result.

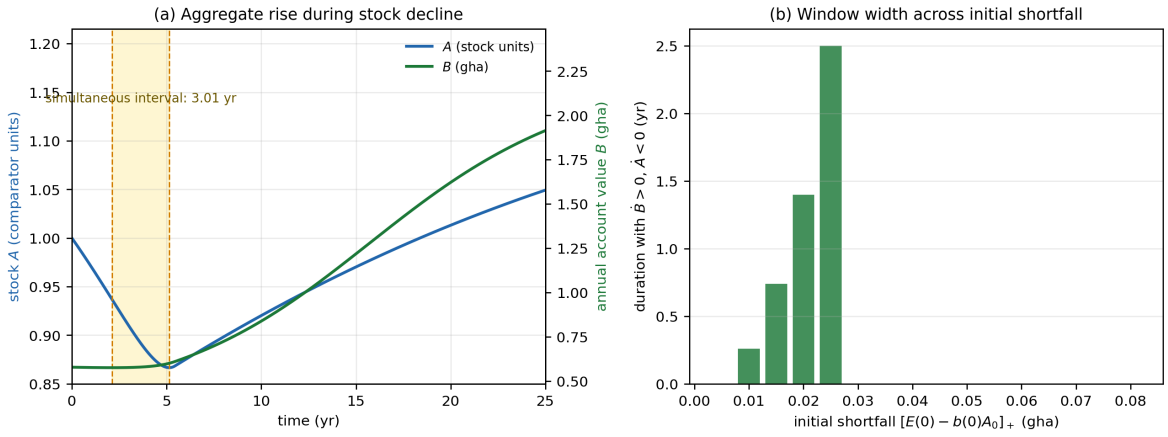


Figure S2: One-stock masking example with fixed demand and no delays. Parameters: $A_0 = 1.0$ stock unit, $E = 0.60\,\mathrm{gha}$, $b_0 = 0.5$, $b_G = 0.8$, $\rho = 0.05\,\mathrm{yr}^{-1}$, $A_{\max} = 1.2$ stock units, $\eta = 0.05\,\mathrm{yr}^{-1}$, $\alpha = 0.03\,\mathrm{gha}^{-1}\,\mathrm{yr}^{-1}$, $\Delta b = 1.5\,\mathrm{gha}$ per stock unit, $\kappa = 0.2\,\mathrm{yr}^{-1}$, and $t_w = 15\,\mathrm{yr}$; $\sigma = 1$, $\omega_{Tb} = 1$, and $D_0 = 0$. Under the exact positive-part operator, the interval with $\dot{B} > 0$ and $\dot{A} < 0$ lasts about 3.0 yr; over it, A falls from about 0.937 to 0.867 stock units while B rises from about 0.575 to 0.600 gha. Panel (b) shows the interval width across initial shortfalls $[E(0) - b(0)A_0]_+$ for this parameterization. These are conditional comparator outputs.

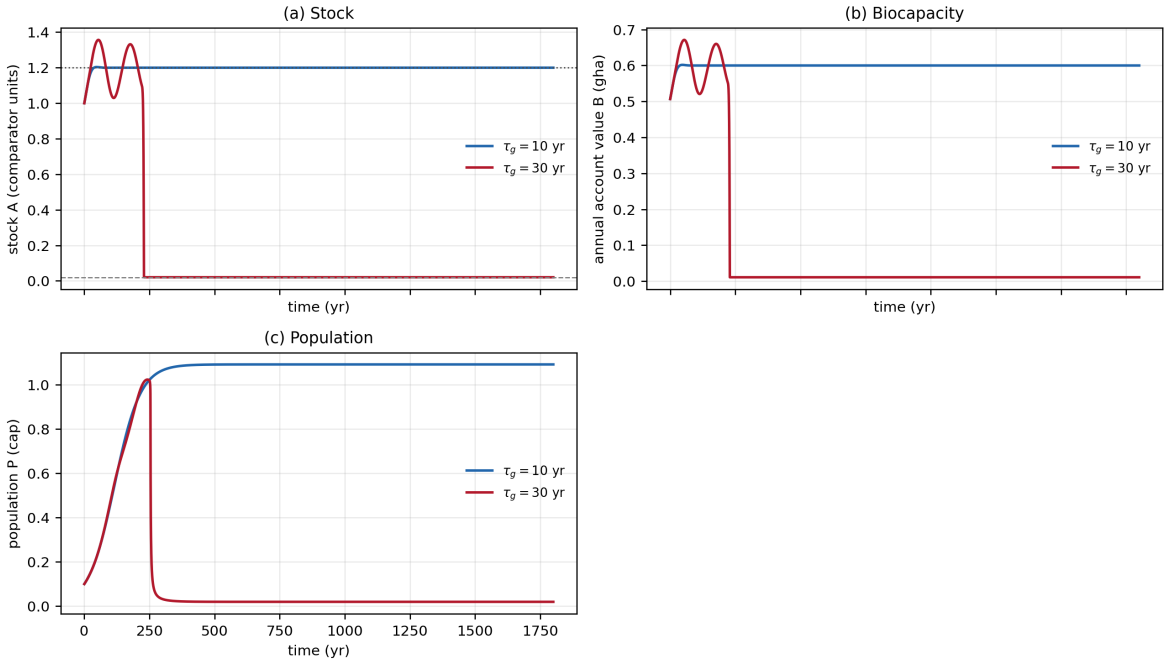


Figure S3: Controlled regeneration-delay examples in the debt-free one-stock subsystem. Both runs use $A_0 = 1.0$ stock unit, $P_0 = 0.1$ cap, $b = 0.5$, $b_G = 0.8$, $\rho = 0.05\,\mathrm{yr}^{-1}$, $A_{\max} = 1.2$ stock units, $e = 0.55$ gha cap^{-1}, $r = 0.02\,\mathrm{yr}^{-1}$, $\tau_p = 25$ yr, $A_{\mathrm{ext}} = 0.02$ stock units, $K_{\min} = 10^{-6}$ cap, and $\sigma = 1$; only τ_g differs (10 versus 30 yr). B is in gha, P in cap, and A in comparator-specific stock units. The selected trajectories do not determine a general critical delay.

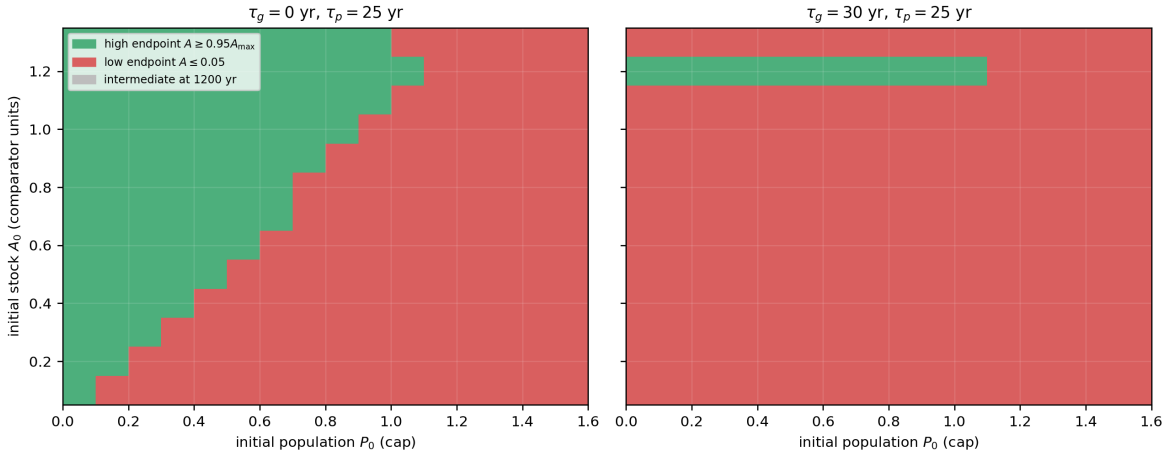


Figure S4: Finite-grid initial-condition classification in the debt-free one-stock subsystem. The debt-free runs use the same parameter set as Fig. S3, with $K_{\min} = 10^{-6}$ cap and $\tau_p = 25$ yr; $\tau_g = 0$ yr in the left panel and 30 yr in the right; the imposed lower bound is $A_{\mathrm{ext}} = 0.02$ stock units. The grid is $A_0 = 0.10, 0.20, \ldots, 1.30$ stock units and $P_0 = 0.05, 0.15, \ldots, 1.55$ cap; trajectories are integrated to 1200 yr with $\Delta t = 0.5$ yr. Green denotes a high endpoint $A(1200) \geq 0.95A_{\max}$, red a low endpoint $A(1200) \leq 0.05$ stock units, and gray an intermediate endpoint. These display-grid classifications are not probabilities or continuous-basin measures.

S8 Controlled technology-wave experiment

The matched experiment compares endogenous-demand runs with T_b active or inactive, using $P_0 = 1.0$ or $1.2\,\mathrm{cap}$. Both runs in each pair share $A_{tot} = 2.5\,\mathrm{ha}$, $A_{f0} = 0.6\,\mathrm{ha}$, $A_{c0} = 0.9\,\mathrm{ha}$, $q_0 = 0.9$, and $D_0 = 0.0\,\mathrm{gha\,yr}$. Parameter values and numerical methods match; only ω_{Tb} differs. The common pre-run histories are constant at $q_0 = 0.9$ and the no-wave initial account value $B = 0.560184\,\mathrm{gha}$ in both runs. Shared parameters are $b_{f0} = 0.85\,\mathrm{gha\,ha^{-1}}$, $b_c = 0.05\,\mathrm{gha\,ha^{-1}}$, $b_{G,c} = 0.8\,\mathrm{gha\,yr\,ha^{-1}}$, $\rho_c = 0.08\,\mathrm{yr^{-1}}$, $\alpha = 0.03\,\mathrm{gha^{-1}\,yr^{-1}}$, $\eta = 0.05\,\mathrm{yr^{-1}}$, $\eta_f = 0.05\,\mathrm{yr^{-1}}$, $\mu = 0.06\,\mathrm{yr^{-1}}$, $e = 0.55\,\mathrm{gha\,cap^{-1}}$, $r = 0.02\,\mathrm{yr^{-1}}$, $\tau_g = 20\,\mathrm{yr}$, $\tau_p = 25\,\mathrm{yr}$, $\tau_{conv} = 1\,\mathrm{yr}$, and $\chi_r = 0.10\,\mathrm{gha^{-1}\,yr^{-1}}$. The active-wave parameters are $\Delta b = 1.0\,\mathrm{gha\,ha^{-1}}$, $\kappa_w = 0.1\,\mathrm{yr^{-1}}$, and $t_{wave} = 40\,\mathrm{yr}$; the integration horizon is $250\,\mathrm{yr}$. The logistic increment is nonzero at $t = 0$ ($T_b(0) = 0.017986\,\mathrm{gha\,ha^{-1}}$), giving $B(0) = 0.560184\,\mathrm{gha}$ without the wave and $0.570976\,\mathrm{gha}$ with it. The difference at $t = 0$ arises from the logistic tail; the land, quality, population, and debt states are otherwise matched, and the pre-run B history remains common. The wave reaches half its increment at $t_{wave} = 40\,\mathrm{yr}$.

Figure S5 shows the $P_0 = 1.0$ pair. Its all-time debt maximum occurs after the wave midpoint in both runs: $D_{max} = 0.008126\,\mathrm{gha\,yr}$ at $86.15\,\mathrm{yr}$ without the wave, versus $0.020205\,\mathrm{gha\,yr}$ at $115.60\,\mathrm{yr}$ with it. At $P_0 = 1.2$, by contrast, the all-time maximum occurs early and is lower with the wave, while the maximum over $t \geq t_{wave}$ is higher. Table S13 reports both all-time maxima and maxima over $t \geq t_{wave}$. Because t_{wave} is the logistic midpoint, the post-midpoint window includes the continued rise of the wave.

Table S13: Matched technology-wave/no-wave debt maxima. Entries are D_{max} in gha yr, with the time of the maximum in years in parentheses; the ratio is wave/no-wave. "All" is the maximum over $0 \leq t \leq 250\,\mathrm{yr}$; $t \geq 40$ is the post-midpoint maximum. The $P_0 = 1.0$ all-time peaks both occur after the midpoint.

P_0 (cap)	quality update	Δt (yr)	window	no wave D_{max} (time)	wave D_{max} (time)	ratio
1.0	delayed	0.050	all	0.008126 (86.15)	0.020205 (115.60)	2.487
1.0	delayed	0.025	all	0.008105 (86.20)	0.020195 (115.65)	2.492
1.0	current	0.050	all	0.008326 (85.70)	0.020168 (115.60)	2.422
1.2	delayed	0.050	all	0.061218 (1.65)	0.051037 (1.45)	0.834
1.2	delayed	0.050	$t \geq 40$	0.010854 (81.70)	0.025184 (112.55)	2.320
1.2	delayed	0.025	all	0.060732 (1.625)	0.050582 (1.475)	0.833
1.2	delayed	0.025	$t \geq 40$	0.010808 (81.775)	0.025165 (112.625)	2.328

For $P_0 = 1.0$, the all-time wave/no-wave debt-peak ratio is 2.487 at $\Delta t = 0.05\,\mathrm{yr}$, 2.492 at $0.025\,\mathrm{yr}$, and 2.422 under the current-quality specification at $0.05\,\mathrm{yr}$. For $P_0 = 1.2$, the corresponding all-time ratios are 0.834 and 0.833, while the post-midpoint ratios are 2.320 and 2.328. Debt maxima change by less than 0.9% under step halving. In the $P_0 = 1.0$ pair, A_c does not fall below its initial value of $0.9\,\mathrm{ha}$; the experiment therefore illustrates a conditional delayed debt rebound, not the capital-land drawdown in the frozen-demand baseline. The $P_0 = 1.2$ contrast shows that the all-time maximum and the post-midpoint maximum are distinct summaries. D is the model's leaky time-integral of shortfall, not an observed NFA or physical-stock variable.

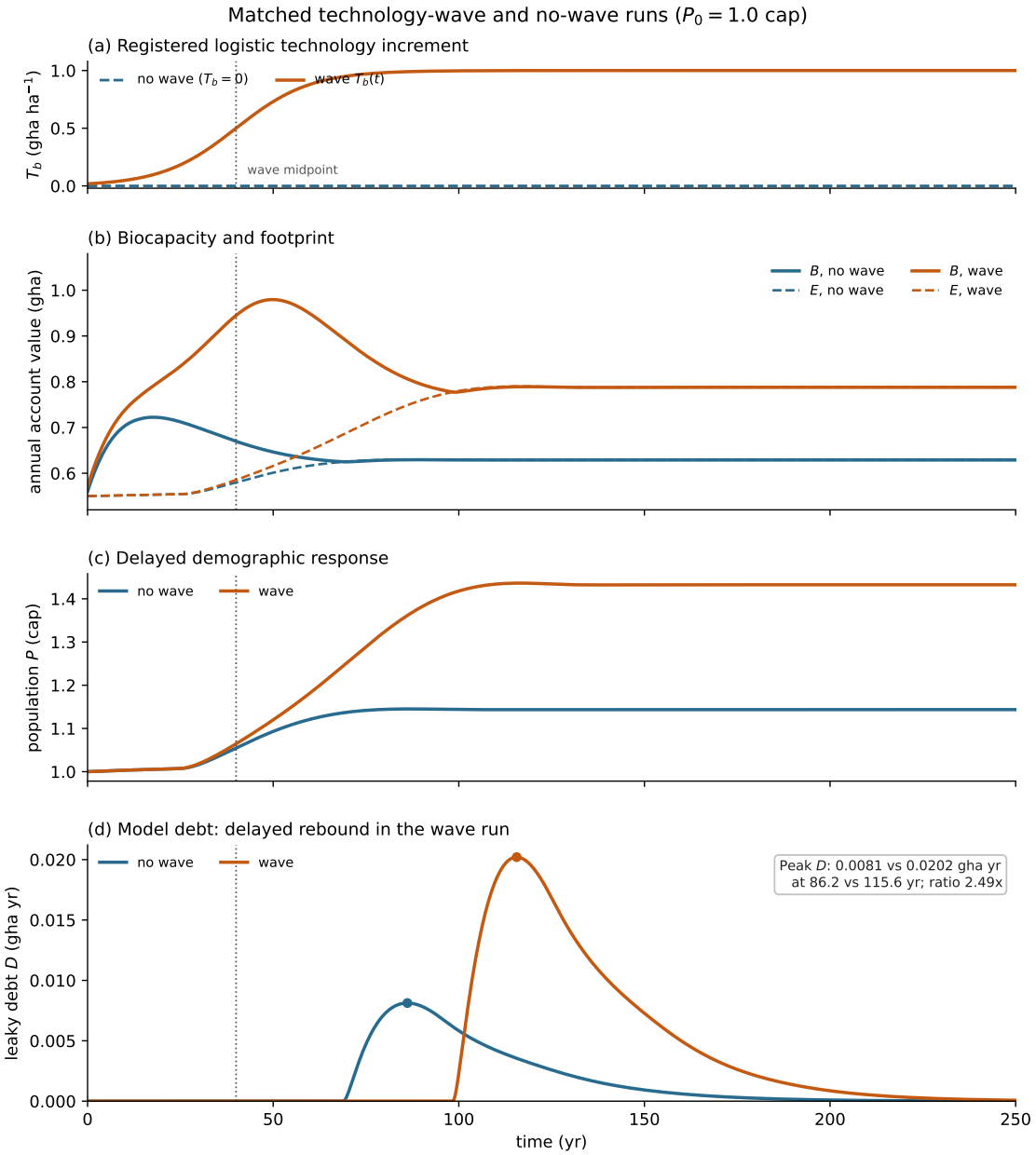


Figure S5: Matched endogenous-demand technology-wave/no-wave simulation at $P_0 = 1.0$ cap. Parameters and initial conditions are listed in SI §S8; only T_b is switched on or off. (a) The active logistic increment $T_b(t)$, with the dotted line at its midpoint $t_{wave} = 40$ yr; (b) annual account values B and E (gha); (c) population P (cap); (d) leaky model debt D (gha yr). Markers denote all-time maxima, which occur at 86.15 yr without the wave and 115.60 yr with it; peak debt is 0.008126 versus 0.020205 gha yr. The plotted wave begins with its small logistic tail at $t = 0$ and approaches $\Delta b = 1.0$ gha ha^{-1}; the dotted line marks the midpoint, not an end date.

Supplementary references

References

Abaee, A. (2026a). *Governance delay and the stability of harvested stocks: mobilising and protective feedback rules, and the review interval as a design parameter*. Working paper. https://github.com/MIKEAA2020/general-sustainability/blob/c1168a898f43348424ae9ea887a79c0def4d6747/arena%20agent%201/paper%20rewrites/latex/paper4_delay_dynamics_v41.pdf.

Abaee, A. (2026b). *Aggregate Indices and Transition Safety: A Quantifier-Order Separation Between Scalarized and Coordinate-Wise Feasibility*. Working paper. https://github.com/MIKEAA2020/general-sustainability/blob/18344877c6b7453e9a60234b60a85fb521d5a998/arena%20agent%201/paper%20rewrites/latex/paper1_assessment_separation_v59.pdf.

Global Footprint Network (2009). *Ecological Footprint Standards 2009*. Global Footprint Network. https://www.footprintnetwork.org/content/uploads/2019/05/Ecological_Footprint_Standards_2009.pdf.

Global Footprint Network (2019). *Working Guidebook to the National Footprint Accounts 2019*. Global Footprint Network. https://www.footprintnetwork.org/content/uploads/2019/05/National_Footprint_Accounts_Guidebook_2019.pdf.

Hutchings, J. A. & Reynolds, J. D. (2004). Marine fish population collapses: consequences for recovery and extinction risk. *BioScience*, 54(4), 297–309.

Lo, K., Basnet, N., Miller, E., Srikanthalingam, B., Foley, B., Van Berkum, J. L., Long, A. H., Toneva, P., Ermina, M. & Klinkenberg, C. (2026). *National Ecological Footprint and Biocapacity Accounts, Edition 2026 (1961–2025): Results and metadata, version 1.0*. Prepared for Footprint Data Foundation by researchers at York University and the University of Iceland. Workbook: https://footprint.info.yorku.ca/files/2026/04/NEFBA_Data_global_national_ecological_footprint_biocapacity.xlsx; release notes: https://footprint.info.yorku.ca/files/2026/04/NEFBA_ReleaseNotes.pdf; data page: https://footprint.info.yorku.ca/data/.

Neubauer, P., Jensen, O. P., Hutchings, J. A. & Baum, J. K. (2013). Resilience and recovery of overexploited marine populations. *Science*, 340(6130), 347–349.

Our World in Data (2026a). *Cropland area*. FAOSTAT data accessed through Our World in Data; grapher page last updated 25 February 2026; accessed 24 October 2026. https://ourworldindata.org/grapher/cropland-area.

Our World in Data (2026b). *Cereal yield*. FAOSTAT data accessed through Our World in Data; grapher page last updated 25 February 2026; accessed 24 October 2026. https://ourworldindata.org/grapher/cereal-yield.

Poeplau, C., et al. (2011). Temporal dynamics of soil organic carbon after land-use change in the temperate zone–carbon response functions as a model approach. *Global Change Biology*, 17(7), 2415–2427.

Poorter, L., et al. (2016). Biomass resilience of Neotropical secondary forests. *Nature*, 530(7590), 211–214.

United Nations et al. (2021). *System of Environmental-Economic Accounting—Ecosystem Accounting (SEEA EA)*. White cover publication, pre-edited text subject to official editing. New York: United Nations. https://seea.un.org/ecosystem-accounting.